SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 30marks
Understanding of Concepts	25	6
Experimental Design	30	7
Use of Tools	20	5
Analysis of Results	15	3
Documentation	10	2

Total Marks 23 **/30**

Signature of the Faculty

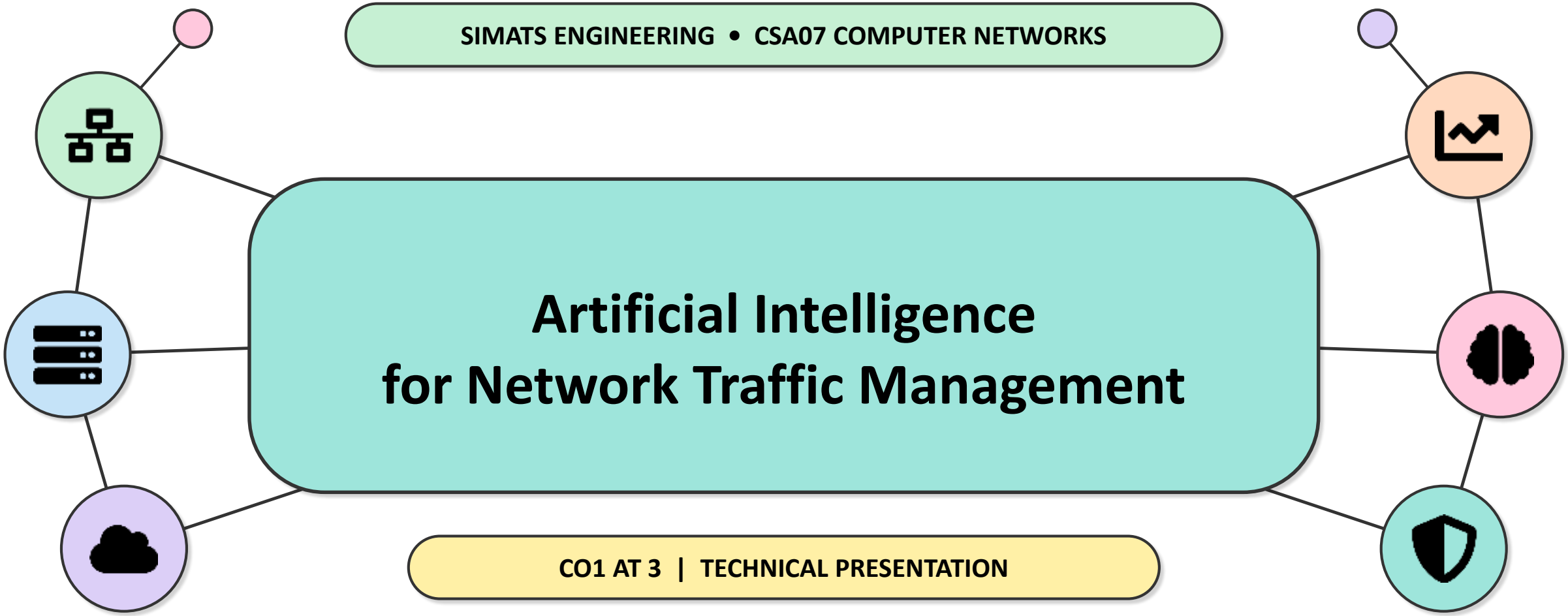
TECHNICAL PRESENTATION QUESTIONS

1. Role of the OSI Model in Software-Defined Networking (SDN)
2. Network Function Virtualization (NFV) and the OSI Architecture
3. Artificial Intelligence for Network Traffic Management
4. Machine Learning-Based Error Detection in Networks
5. Digital Twin Networks and the OSI Model
6. 6G Communication Architecture Compared with the OSI Model
7. Edge Computing and Reliable Data Transmission
8. Cloud Networking Through the OSI Architecture
9. Satellite Internet Communication Using OSI Layers
10. Future of Reliable Networking Beyond the Traditional OSI Model

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable	Basic analysis with limited insights	Poor or incorrect analysis of results

SIMATS ENGINEERING • CSA07 COMPUTER NETWORKS



Artificial Intelligence for Network Traffic Management

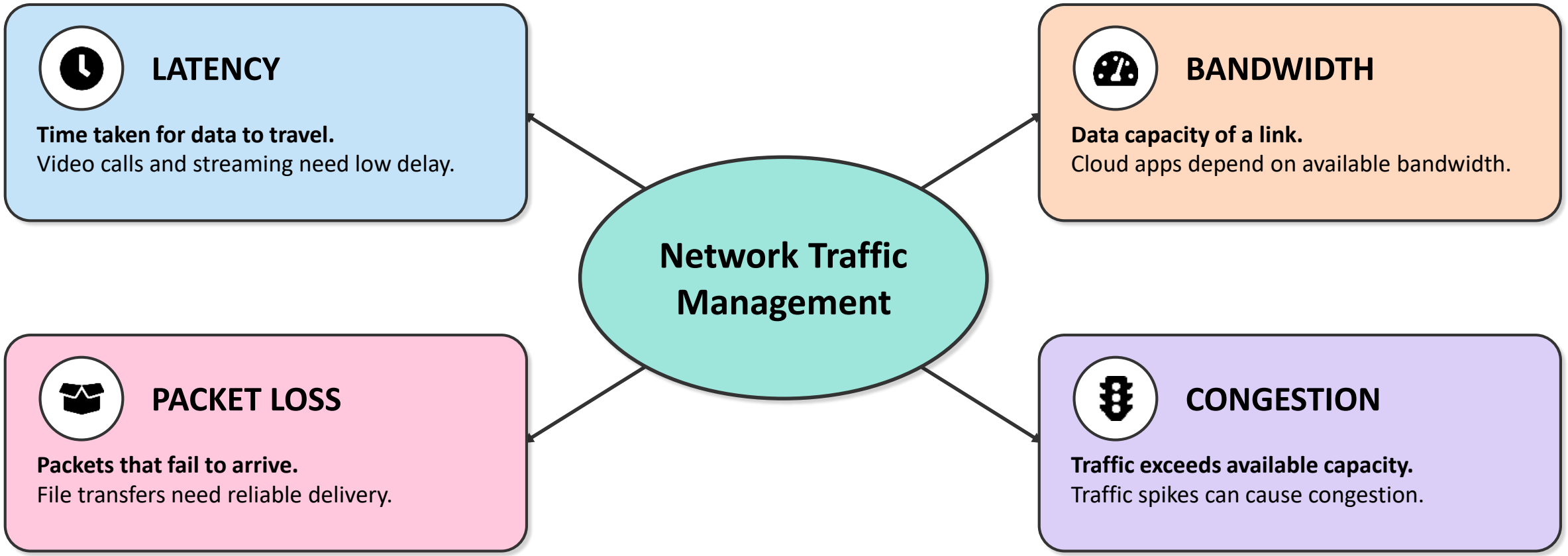
CO1 AT 3 | TECHNICAL PRESENTATION

Name: C P ELTON PEJO

Register No : 1965260136

Why network traffic management matters

Modern networks carry many types of traffic at once.



Goal: deliver data at the right time over a reliable path.

Where AI fits in the OSI architecture

7	**APPLICATION**	User-facing services
6	**PRESENTATION**	Data format and encryption
5	**SESSION**	Session coordination
4	**TRANSPORT**	TCP/UDP; end-to-end delivery
3	**NETWORK**	IP routing and path selection
2	**DATA LINK**	Frames and local delivery
1	**PHYSICAL**	Signals over transmission media

AI ACROSS THE LAYERS

OBSERVE
Collect traffic flows, delay, packet loss and utilization.

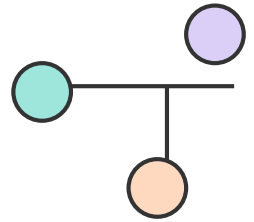
PREDICT
Models estimate traffic demand or flag unusual patterns.

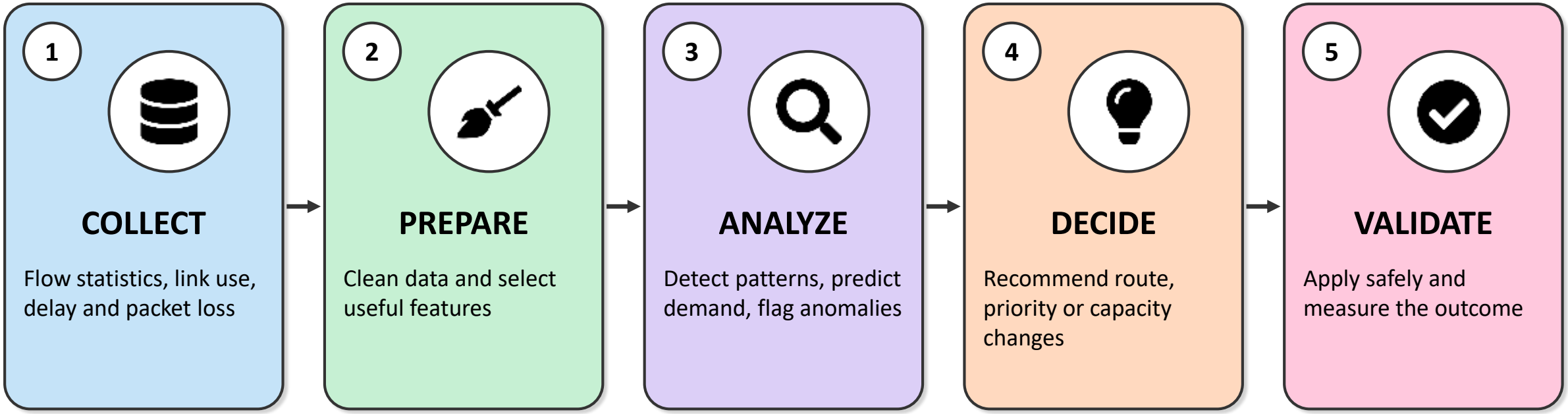
ACT
Recommend changes to routes, queues or capacity.

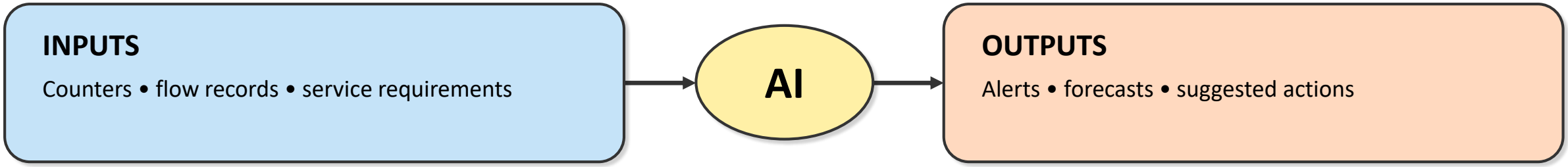
AI is not an OSI layer; it supports network management across layers.

HOW IT WORKS

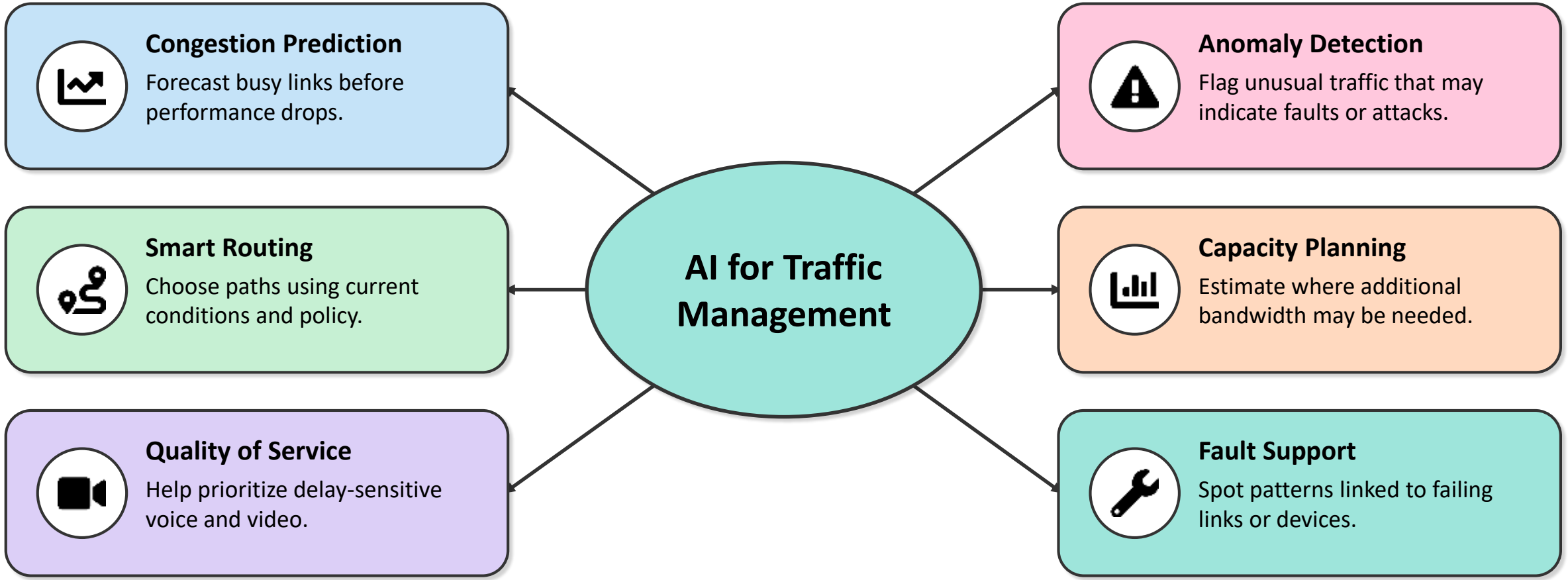
AI-based traffic management workflow







Practical applications



Results depend on data quality, model design and network policy.

EXPERIMENTAL DESIGN

Tools and a simple experiment

SIMULATION

Mininet or a comparable network emulator.

CONTROLLER

Ryu / OpenFlow-compatible SDN controller (optional).

ANALYSIS

Python, pandas and scikit-learn.

PROPOSED EXPERIMENT

1. Generate normal and bursty traffic in a test topology.
2. Record throughput, delay, packet loss and link use.
3. Compare fixed routing with AI-assisted recommendations.

FAIR COMPARISON

Use the same topology and traffic for both cases. Repeat runs, record settings and compare averages. No improvement is claimed until measurements are collected.

This is a proposed design; no experimental results are claimed.

ANALYSIS

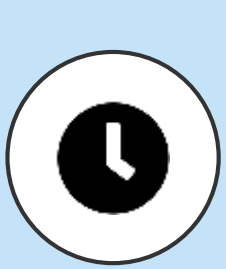
How performance should be evaluated



THROUGHPUT

Higher is usually better

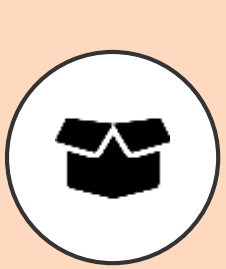
Useful data delivered per second



LATENCY

Lower is usually better

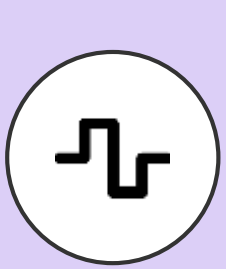
Time for packets to travel



PACKET LOSS

Lower is usually better

Share of packets not delivered



JITTER

Lower is usually better

Variation in packet delay

Compare baseline and AI-assisted measurements; explain trade-offs, not just averages.

Benefits and challenges

BENEFITS

- Earlier congestion warnings
- Faster pattern detection
- Better-informed routing and QoS decisions
- Potentially more efficient capacity use

VS

CHALLENGES

- Incomplete or noisy monitoring data
- Models may drift as traffic changes
- False alarms or poor recommendations
- Privacy, security and explainability concerns

Best practice: human oversight, clear policies and a safe fallback.

TAKEAWAYS

Conclusion

The OSI model helps organize where traffic problems occur.

AI can analyze telemetry, predict demand and support decisions.

AI can make network management more proactive, but it does not replace networking fundamentals.

Controlled experiments should validate any claimed benefit.

Reliable networks still need security, policy and human oversight.

Key message: measure first, automate carefully, verify continuously.

DISCUSSION

References & questions

Background reading

1. **ISO/IEC 7498-1** — OSI basic reference model.
2. **IETF RFC 791** — Internet Protocol (IPv4).
3. **Open Networking Foundation** — SDN resources.
4. **Mininet documentation** — network emulation.
5. **scikit-learn documentation** — machine-learning model evaluation.

Thank You!

Questions?

SIMATS ENGINEERING

SIMATS

Artificial Intelligence
for Network Traffic Management

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GPS Map Camera

Mevalurkuppam, Tamil Nadu, India 🇮🇳
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